

Alexander Brown

San Jose, CA | alexjirocoll@gmail.com | 408.355.5966 | linkedin.com/in/alexander-jiro-brown/ | github.com/alexajb2

PROFESSIONAL SUMMARY

Computer Science M.S. candidate at Georgia Tech specializing in artificial intelligence, with experience building and evaluating AI systems across computer vision, reinforcement learning, and time-series modeling. Graduate researcher applying object detection to ecological camera-trap footage, with a background in full-stack development and production software supporting 9,000+ users.

EDUCATION

Georgia Institute of Technology

August 2025 - Expected May 2027

M.S. Computer Science, Artificial Intelligence Specialization

Relevant Coursework: Machine Learning, Deep Learning, Robotics: AI Techniques, Game AI, Introduction to Health Informatics

GPA: 4.0

University of California, Irvine

September 2022 - June 2025

B.S. Computer Science, Intelligent Systems Specialization

Relevant Coursework: Artificial Intelligence, Machine Learning and Data Mining, Graphical Models, Information Retrieval, Algorithm Analysis, Graph Algorithms, Embedded Software, Probability and Statistics

GPA: 3.85

RESEARCH

Graduate Research Assistant

May 2026 - Present

Georgia Tech Human Augmented Analytics Group (HAAG)

- Train and evaluate object detection models including YOLO, RT-DETR, DETR, and Swin Transformer-based architectures to detect and classify Hume's Leaf Warblers from Himalayan camera-trap footage
- Benchmark detection performance on 50,000+ annotated frames, including 13,000+ positive examples, to support model selection for annotation-limited ecological computer vision data
- Preparing a first-author journal manuscript on automated wildlife detection and behavior analysis, with draft targeted for August 2026 and submission/publication target by December 2026

Machine Learning Research Assistant

September 2024 - April 2025

California Institute of Telecommunications & Information Technology (Cal IT2)

- Developed GRU, LSTM, RNN, and RBF-RNN time-series models in PyTorch to simulate thermal energy storage system behavior from sensor data, reducing inference time to under 1 second vs. a 10-minute traditional simulation runtime

WORK EXPERIENCE

Full-Stack Web Developer

March 2023 - June 2025

University of California, Irvine, Student Affairs Information Technology

- Developed and maintained full-stack web applications using Laravel, PHP, JavaScript, CSS, and MySQL, supporting 9,000+ annual users across student-facing UCI systems
- Automated QA workflows using Cypress testing and custom scripts, improving release reliability

PROJECTS

AI BWC Footage Tagging and Purging

January 2025 - June 2025

Capstone Project Team Lead

- Awarded 1st Place Computer Science Winner at UCI ICS Expo
- Built AI-driven computer vision software for the Signal Hill Police Department to identify and categorize body-worn camera footage
- Implemented ensemble ML models incorporating ImageBind, EfficientNet, and transformer architectures to classify 16 terabytes of untagged footage

Reinforcement Learning Agent: Cuphead

January 2025 - March 2025

AI Project

- Built a reinforcement learning agent to defeat Cuphead's Root Pack boss by combining YOLO-based real-time object detection with a Deep Q-Network (DQN) in a fast-paced, visually complex environment

SKILLS

Languages & Tools: Python, SQL, Git, Docker, AWS, Claude Code, GPT Codex, AI coding agents, Bash, Java, C++

Frameworks & Libraries: PyTorch, Hugging Face, Scikit-learn, NumPy, pandas, Matplotlib

Techniques: Computer Vision, Machine Learning, Reinforcement Learning, CI/CD, End-to-End Testing